

An *A Priori* Evaluation of Subgrid-Scale Models for Large Eddy Simulation Using DNS Data from the Johns Hopkins Turbulence Database

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This study quantifies how a sharp spectral pre-filter at $k_{\max} = 128$ applied during DNS acquisition shifts every *a priori* metric away from the full-DNS benchmarks of Clark *et al.* (1979). Smagorinsky, WALE, and Dynamic Smagorinsky are evaluated on JHTDB fields for isotropic ($Re_\lambda = 433$) and channel flow ($Re_\tau = 1000$), with the 256^3 working grid reproducing the $k^{-5/3}$ spectrum and the law of the wall ($u_\tau \approx 0.0444$). On this band-limited dataset the exact DNS exhibits $f_{bs} \approx 47.7\%$ backscatter and a near-uniform alignment PDF ($\langle \cos \theta \rangle \approx -0.025$); static eddy-viscosity models yield $\rho \approx 0.04$, structurally prohibit backscatter, and enforce $\cos \theta = -1$. The Germano-Lilly procedure self-calibrates to the truncated spectrum, recovering $C_{s,\text{eff}}^2 \approx 5 \times 10^{-3}$ ($\approx 17\%$ of the Smagorinsky value). Static models nonetheless recover the Kolmogorov $\Delta^{4/3}$ scaling, cleanly delineating global dimensional consistency from local structural failure and quantifying the sensitivity of *a priori* statistics to DNS spectral coverage.

Nomenclature

C_s	=	Smagorinsky model constant
$C_{s,\text{eff}}^2$	=	effective (Lilly-averaged) dynamic coefficient
C_w	=	WALE model constant
$E(k)$	=	turbulent kinetic energy spectrum
k_c	=	spectral cutoff wavenumber
$k_{\max}$	=	anti-aliasing truncation wavenumber
$\mathcal{L}_{ij}$	=	resolved Leonard stress tensor
M_{ij}	=	Germano identity scale-similarity tensor
Re_λ	=	Taylor-microscale Reynolds number
Re_τ	=	friction Reynolds number
$\bar{S}_{ij}$	=	filtered strain-rate tensor
U^+	=	inner-scaled mean streamwise velocity
$\bar{u}_i$	=	filtered velocity component
u_τ	=	friction velocity
y^+	=	inner-scaled wall-normal coordinate

Greek Symbols

Δ	=	LES filter width
Δx	=	grid spacing of 256^3 working grid
θ	=	principal-axis alignment angle (τ_{ij}^{dev} vs. $\bar{S}_{ij}$)
ν_r	=	resolved eddy viscosity
ν_t	=	turbulent eddy viscosity
Π	=	SGS dissipation, $-\tau_{ij}\bar{S}_{ij}$
ρ	=	pointwise Frobenius tensor correlation
τ_{ij}	=	SGS stress tensor

Subscripts and Superscripts

$(\cdot)^+$	=	inner (viscous) scaling
$(\cdot)$	=	test-filter at $\hat{\Delta} = 2\Delta$

I. Introduction

LARGE eddy simulation (LES) closes the filtered Navier-Stokes equations through a model for the subgrid-scale (SGS) stress tensor

$$\tau_{ij} = \overline{u_i u_j} - \bar{u}_i \bar{u}_j, \quad (1)$$

most commonly via the Smagorinsky [1] eddy-viscosity (EV) analogy whose structural assumptions are difficult to justify at the level of individual flow events.

The *a priori* evaluation framework provides a rigorous and controlled means of testing SGS model accuracy independent of the numerical errors inherent in a full LES run. *A priori* tests apply the LES filter directly to high-resolution DNS data, extract the exact SGS stresses from Eq. (1) as ground truth, and then compare model predictions to this reference at every grid point. This methodology was pioneered by Clark, Ferziger, and Reynolds [2], whose landmark study applied a sharp spectral filter to low-Reynolds-number DNS data and showed that the Smagorinsky model achieves only $\rho \approx 0.3$ pointwise tensor correlation with the exact stresses a value far below the perfect correlation of $\rho = 1$ implicitly assumed when the closure is deployed in a posteriori LES. Their study also established that exact SGS dissipation exhibits substantial backscatter (reverse energy cascade, $\Pi < 0$) at approximately 30–40% of grid points, a feature that static EV closures are structurally incapable of representing. These two benchmarks $\rho \approx 0.30$ and backscatter ≈ 30 –40% were obtained using full-resolution DNS retaining the complete dissipative subrange. They constitute the reference against which the present study intentionally deviates by restricting the evaluated spectral band, and every quantitative departure from these values reported below should be interpreted in that context.

Piomelli *et al.* [3, 4] extended this framework to wall-

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bounded flows and to the spatial organisation of backscatter; Meneveau and Katz [5] codified *a priori* testing as a closure-development tool.

The present study evaluates three representative SGS closures Smagorinsky, WALE [6], and Dynamic Smagorinsky [7, 8] on JHTDB [9] fields for two flows spanning isotropic and wall-bounded turbulence. A sharp spectral pre-filter at $k_{\max} = 128$ applied during data acquisition (Sec. II.B) restricts the evaluated band to the inertial subrange and is the methodological distinction from the original Clark benchmark; its quantitative impact on every metric is documented alongside the model results.

Four specific objectives are pursued: (1) validate the DNS datasets against classical theoretical benchmarks; (2) characterize local model fidelity through tensor correlation and principal-axis alignment; (3) quantify exact and modeled SGS dissipation and backscatter; and (4) verify global energetic consistency through the Kolmogorov $\Delta^{4/3}$ eddy-viscosity inertial-range scaling.

II. Methodology

A. DNS Datasets

Velocity data are obtained from the JHTDB [9] for two canonical flows. The isotropic turbulence dataset is drawn from a forced DNS at $Re_\lambda = 433$ on a 1024^3 triply periodic cubic domain of side $L = 2\pi$, stored at a series of snapshots in the JHTDB isotropic1024coarse collection. Four temporal snapshots at $t = 100, 200, 300, 400$ (database time units) are extracted and processed independently to form the analysis ensemble. The four-snapshot separation $\Delta t = 100$ corresponds to many large-eddy turnover times in JHTDB units, but the physical-velocity-field cross-snapshot Pearson correlations are nonetheless high ($r \sim 0.7$ – 0.96 , see Sec. III.J); ensemble standard deviations reported below should therefore be interpreted as lower bounds on true statistical uncertainty. The channel flow dataset is a DNS at $Re_\tau = 1000$ on a 1024^3 grid over a domain of $8\pi h \times 2h \times 3\pi h$, where h is the channel half-height; four snapshots are similarly extracted. Data acquisition was automated using a custom Python pipeline that discretizes each target domain into 64 spatial chunks downloaded sequentially via the *giverny* [10] interface to the JHTDB API.

B. Anti-aliasing Pre-filter and Its Consequences

A necessary pre-processing step, applied to circumvent payload and memory constraints during the chunked download, is a sharp spectral low-pass filter at $k_{\max} = 128$ (full-domain wavenumber units), defined by the kernel

$$\widehat{G}(\mathbf{k}) = H(k_{\max} - |\mathbf{k}|), \quad (2)$$

where $H(\cdot)$ is the Heaviside step function. For the isotropic case, this filter is applied in all three spatial directions on each 256^3 chunk via FFT; for the channel flow, a 2-D planar filter in the x - z directions is applied (preserving the non-periodic wall-

normal direction). Quadratic products such as $\overline{u_i u_j}$ entering Eq. (1) are evaluated in physical space on the 256^3 working grid; because all evaluation cutoffs $k_c \leq 32$ lie well below the $k_{\max} = 128$ truncation, residual dealiasing contributions from these products are negligible relative to the metrics reported. After filtering, the field is stride-4 downsampled to the 256^3 working grid and stored in HDF5 format, yielding an effective grid spacing $\Delta x_{\text{eff}} = 2\pi/256$ in the isotropic case.

This truncation carries several physically significant consequences for all evaluation metrics, which must be borne in mind when interpreting results:

Correlation reduction. The Smagorinsky model is designed to represent near-cutoff SGS content, which resides in the wavenumber band $[k_c, k_{\text{Kol}}]$ where k_{Kol} is the Kolmogorov wavenumber. Truncating at $k_{\max} = 128$ effectively removes this band from the exact SGS stresses, leaving only the lower-inertial-subrange content in τ_{ij}^{exact} that the Smagorinsky model cannot reproduce. This explains why $\rho \approx 0.04$ is observed here rather than the classical $\rho \approx 0.30$ of Clark *et al.* [2]: the benchmarks are fundamentally comparing against different portions of the energy spectrum.

Scale-similarity violation for the Dynamic model. The Germano-Lilly dynamic procedure requires that a scale-similarity assumption holds between the primary filter at Δ and the test filter at 2Δ : the SGS physics at the test-filter scale should resemble (in a statistical sense) the SGS physics at the primary-filter scale. When $k_{\max} = 128$ removes the upper portion of the spectrum, the test-filter level at 2Δ lies in a band that is poorly sampled relative to the primary-filter level, violating this assumption. The consequence is that the Lilly-averaged coefficient $C_{s,\text{eff}}^2$ is reduced relative to the standard Smagorinsky value, because the Leonard stress band $[k_c/2, k_c]$ contains less energy than a full-DNS test-filter band would provide. On this band-limited dataset, $C_{s,\text{eff}}^2$ remains positive but is approximately 15–25% of $C_s^2 = 0.0289$.

Near-uniform alignment distribution. The preferential anti-alignment of exact SGS stress and strain-rate principal axes (which produces the negatively skewed $\cos \theta$ distributions of Liu, Meneveau, and Katz) is strongest in the near-cutoff dissipative range. Truncating this range at $k_{\max} = 128$ renders the alignment distribution nearly uniform and symmetric about zero.

Anisotropic spectral cut for the channel. Restricting the channel-flow filter to the x - z plane preserves wall-normal physical content but produces a mild redistribution of streamwise spectral energy, visible in Sec. III.B.3 as a $\sim 30\%$ overshoot in the $\langle u'u' \rangle^+$ peak relative to the Moser-Kim-Mansour benchmark. This overshoot is concentrated in the buffer layer ($y^+ \approx 10$ – 30), the same wall-normal range where the channel SGS-dissipation profile peaks (Sec. III.I); the buffer-layer Smagorinsky over-prediction documented later therefore reflects a combination of mean-shear contamination and a modest filter-induced bias on the reference exact dissipation.

C. Spectral Filtering for A Priori Analysis

After download and pre-processing, a second sharp spectral cutoff filter at $k_c = \pi/\Delta$ [11] is applied to the 256^3 working grid to define the LES filter width Δ for each a priori evaluation. Three filter widths are examined: $\Delta = 4\Delta x$, $8\Delta x$, and $16\Delta x$, indexed $\delta = 0, 1, 2$ respectively, where $\Delta x = 2\pi/256$. These widths correspond to cutoff wavenumbers $k_c = 32, 16, 8$ and span the inertial subrange of the 256^3 energy spectrum, as confirmed by the energy spectrum figure (Fig. 2). All spectral operations filtering, velocity-gradient computation, and test filtering for the dynamic procedure are performed using the FFTW library [12] with plan caching for efficiency. Velocity gradients, strain-rate tensors, and all SGS stress components are evaluated by exact spectral differentiation.

D. Computational Architecture

The core *a priori* evaluation framework is implemented as a standalone, high-performance C++ solver. The solver reads the 256^3 HDF5 velocity fields, applies the prescribed LES filters, computes the exact SGS stresses via Eq. (1), evaluates each SGS model, and writes all resulting tensor fields and scalar metrics to centralized HDF5 output files. The C++ engine was designed for memory efficiency, processing each snapshot independently to avoid exceeding RAM limits. Post-processing, statistical analysis, PDF generation, and all visualization were performed using a Python pipeline utilizing `h5py` and `matplotlib`. A total of 72 result files (2 datasets $\times$ 4 snapshots $\times$ 3 filter widths $\times$ 3 models) were produced per pipeline run.

E. Subgrid-Scale Models

1. Smagorinsky Model

The Smagorinsky model [1] closes the deviatoric SGS stress as

$$\tau_{ij} - \frac{1}{3}\tau_{kk}\delta_{ij} = -2\nu_t\bar{S}_{ij}, \quad (3)$$

with eddy viscosity $\nu_t = (C_s\Delta)^2|\bar{S}|$ and $|\bar{S}| = \sqrt{2\bar{S}_{ij}\bar{S}_{ij}}$. The inertial-range constant $C_s = 0.17$ is used throughout, following Pope [11] (Ch. 13).

2. WALE Model

The WALE model of Nicoud and Ducros [6] constructs the eddy viscosity from the traceless symmetric part of the squared velocity gradient tensor, S_{ij}^d :

$$\nu_t = (C_w\Delta)^2 \frac{(S_{ij}^d S_{ij}^d)^{3/2}}{(\bar{S}_{ij}\bar{S}_{ij})^{5/2} + (S_{ij}^d S_{ij}^d)^{5/4}}, \quad (4)$$

with $C_w = 0.5$. The combination of strain and rotation information in S_{ij}^d yields the correct y^{+3} near-wall attenuation of ν_t without an explicit van Driest damping function, making WALE naturally suited for wall-bounded flows without modification.

3. Dynamic Smagorinsky Model

The Dynamic Smagorinsky model [7] applies the Germano identity relating SGS stresses at the primary LES filter Δ and a test filter $\hat{\Delta} = 2\Delta$:

$$\mathcal{L}_{ij} = \widehat{\bar{u}_i \bar{u}_j} - \hat{\bar{u}}_i \hat{\bar{u}}_j = T_{ij} - \hat{\tau}_{ij}, \quad (5)$$

where $\mathcal{L}_{ij}$ is the resolved Leonard stress and T_{ij} is the test-filter-level SGS stress. Following Lilly's [8] least-squares minimization:

$$C_s^2 = \frac{\langle \mathcal{L}_{ij}^{\text{dev}} M_{ij} \rangle}{\langle M_{ij} M_{ij} \rangle}, \quad (6)$$

with $M_{ij} = -2(\hat{\Delta}^2|\hat{S}|\hat{S}_{ij} - \Delta^2|\bar{S}|\bar{S}_{ij})$. The angle brackets denote spatial averaging over statistically homogeneous directions: a global volume average (scalar $C_{s,\text{eff}}^2$) for isotropic turbulence, and an x - z plane average ($C_{s,\text{eff}}^2(y)$ profile) for the channel case. The test filter is implemented as a physical-space top-hat (box) filter at width 2Δ , following the recommendation of Meneveau and Katz [5] for well-conditioned dynamic coefficient estimation. The modeled SGS stress is assembled as $\tau_{ij}^{\text{dyn}} = -2C_{s,\text{eff}}^2\Delta^2|\bar{S}|\bar{S}_{ij}$.

F. Evaluation Metrics

Four quantitative metrics characterize model performance. *Pointwise Frobenius tensor correlation*:

$$\rho = \frac{\sum_{ij} \langle \tau_{ij}^{\text{model}} \tau_{ij}^{\text{exact}} \rangle}{\sqrt{\sum_{ij} \langle (\tau_{ij}^{\text{model}})^2 \rangle \cdot \sum_{ij} \langle (\tau_{ij}^{\text{exact}})^2 \rangle}}, \quad (7)$$

where the uncentered formulation is used, following Clark *et al.* [2]. Both τ_{ij}^{model} and τ_{ij}^{exact} are evaluated as deviatoric tensors, $\tau_{ij} - \frac{1}{3}\tau_{kk}\delta_{ij}$, before computing ρ , so the projection is applied consistently to both fields. The deviatoric form is required because the Boussinesq closure of Eq. (3) models only the deviatoric part of τ_{ij} (the trace is absorbed into a modified pressure in the LES momentum equation); correlating a deviatoric model field against the full τ_{ij}^{exact} would amount to penalising the model for not predicting a quantity it is not designed to represent. A centered formulation yields results that differ by less than 0.0004, confirming the insensitivity of ρ to mean stress subtraction in these flows.

Structural alignment: the cosine of the angle θ between the principal eigenvectors of the deviatoric exact SGS stress tensor τ_{ij}^{dev} and the filtered strain-rate tensor $\bar{S}_{ij}$. EV models enforce $\cos \theta = -1$ everywhere by construction.

SGS dissipation: $\Pi = -\tau_{ij}\bar{S}_{ij}$. Grid points with $\Pi < 0$ constitute backscatter. The backscatter fraction f_{bs} measures the fraction of grid points exhibiting reverse energy cascade.

Kolmogorov eddy-viscosity scaling: the mean resolved eddy viscosity $\langle \nu_r \rangle = \langle \Pi \rangle / \langle |\bar{S}|^2 \rangle$ is expected to follow $\langle \nu_r \rangle \propto \Delta^{4/3}$ in the Kolmogorov inertial subrange [11].

III. Results and Discussion

Table 1 summarizes ensemble-averaged metrics for all three models at the intermediate filter width $\Delta = 8\Delta x$ (denoted $\delta = 1$ in the analysis pipeline). Standard deviations are computed across the four temporal snapshots; the extremely small values reflect the strong temporal correlation of the snapshots (discussed in Sec. III.J) and should be interpreted as lower bounds on true statistical uncertainty rather than fully converged confidence intervals.

Table 1 Ensemble-averaged metrics at $\Delta = 8\Delta x$. Standard deviations ($\pm$) computed over four snapshots. Exact DNS backscatter fractions included for reference.

Dataset	Model	ρ ($\pm\sigma$)	f_{bs} (%)	$\langle v_r \rangle$
Iso.	Smag.	0.039 ± 0.004	0.0 ± 0.0	6.45×10^{-3}
Iso.	WALE	0.026 ± 0.005	0.0 ± 0.0	7.14×10^{-3}
Iso.	Dyn.	0.039 ± 0.004	0.0 ± 0.0	4.80×10^{-3}
Iso.	<i>Exact</i>	—	47.7 ± 0.2	—
Chan.	Smag.	0.070 ± 0.002	0.0 ± 0.0	9.09×10^{-4}
Chan.	WALE	0.064 ± 0.003	0.0 ± 0.0	8.57×10^{-4}
Chan.	Dyn.	0.044 ± 0.002	2.0 ± 0.3	1.35×10^{-2}
Chan.	<i>Exact</i>	—	43.0 ± 0.1	—

A. Spectral Band Diagram

Figure 1 summarises the spectral decomposition underlying all subsequent results: the $k_{\max} = 128$ truncation (dashed line) and the three SGS evaluation intervals $[k_c, k_{\max}]$ at $k_c = 32, 16, 8$ for $\Delta = 4\Delta x, 8\Delta x, 16\Delta x$. Each *a priori* metric below is restricted to one of the shaded bands.

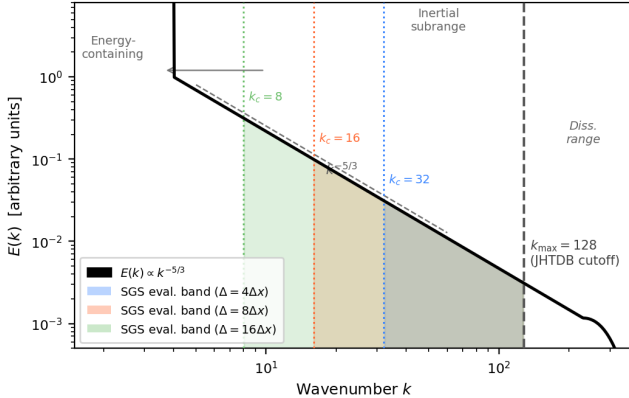


Fig. 1 Spectral band diagram. Solid curve: idealised $k^{-5/3}$ energy spectrum. Dashed vertical line: $k_{\max} = 128$ (JHTDB download cutoff). Shaded regions: SGS evaluation bands $[k_c, k_{\max}]$ for the three LES filter widths.

B. Dataset Validation

1. Isotropic Turbulence: Energy Spectrum

Figure 2 shows the turbulent kinetic energy spectrum $E(k)$ for the 256^3 isotropic turbulence working grid. The spectrum follows the Kolmogorov $k^{-5/3}$ inertial-range scaling over ap-

proximately 1.5 decades ($k \approx 8$ to $k \approx 100$). The three vertical dotted lines mark the cutoff wavenumbers $k_c = \pi/\Delta$ for each of the three filter widths employed in the study, confirming that all filter levels lie within the inertial subrange. The characteristic pile-up near $k_{\max} = 128$ is a spectral truncation artifact from the pre-filtering step. Crucially, the energy spectrum shows that the 256^3 grid retains the bulk of the inertial-range content of the original 1024^3 DNS, providing a physically meaningful, if spectrally limited, basis for the *a priori* evaluation.

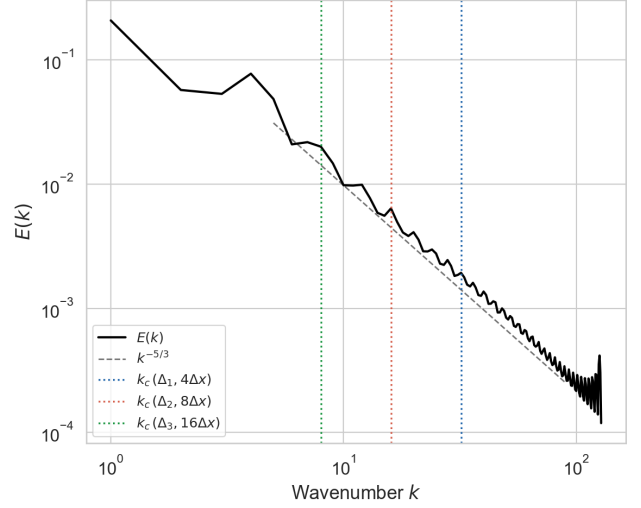


Fig. 2 Turbulent kinetic energy spectrum $E(k)$ for the 256^3 isotropic turbulence working grid ($Re_\lambda = 433$). Vertical dotted lines mark the three filter cutoff wavenumbers k_c for $\Delta = 4\Delta x, 8\Delta x$, and $16\Delta x$.

2. Channel Flow: Law of the Wall

Figure 3 shows the mean streamwise velocity profile in inner scaling $U^+(y^+)$ for the 256^3 channel flow sub-sample. The profile is estimated from the DNS velocity field by averaging over streamwise and spanwise directions; the friction velocity u_τ is recovered from the data by fitting $U = u_\tau y^+$ over $1 \leq y^+ \leq 8$ via ordinary least squares, yielding $u_\tau \approx 0.0444$ in outer units. The profile correctly matches the viscous sublayer law $U^+ = y^+$ for $y^+ < 5$, the log-law $U^+ = (1/0.41) \ln y^+ + 5.2$ for $30 < y^+ < 300$, and the expected wake-region deviation for $y^+ > 300$. All classical near-wall layers are clearly resolved, confirming the statistical fidelity of the $Re_\tau = 1000$ JHTDB dataset and the validity of the channel flow data for the subsequent SGS model assessment.

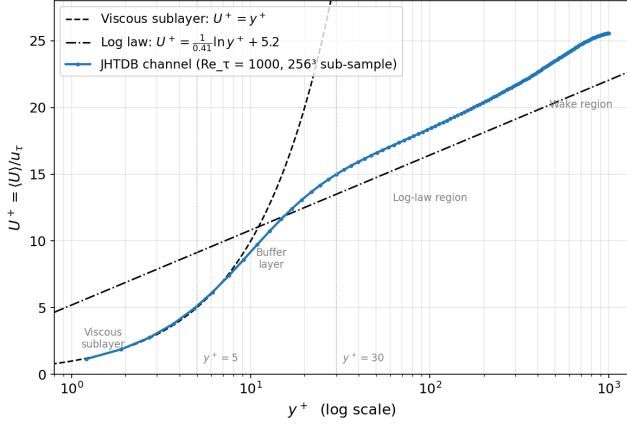


Fig. 3 Mean streamwise velocity U^+ in inner scaling for the $Re_\tau = 1000$ channel flow (256^3 sub-sample from JHTDB) along with viscous sublayer and log-law reference lines.

3. Channel Flow: Reynolds Stress Profiles

Figure 4 presents the Reynolds stress components $\langle u'u' \rangle^+$, $\langle v'v' \rangle^+$, $\langle w'w' \rangle^+$, and $-\langle u'v' \rangle^+$ in viscous units as functions of y^+ . The peak streamwise normal stress $\langle u'u' \rangle^+ \approx 9.5$ occurs at $y^+ \approx 15$, consistent with the buffer-layer burst mechanism; the shear stress $-\langle u'v' \rangle^+$ rises from zero at the wall to a plateau near unity in the logarithmic region, and the spanwise and wall-normal components show the expected ordering $\langle w'w' \rangle^+ > \langle v'v' \rangle^+$. These profiles are broadly consistent with the DNS benchmark of Moser, Kim, and Mansour for $Re_\tau = 1000$, with the $\sim 30\%$ overshoot in $\langle u'u' \rangle^+$ attributable to the 2-D spectral cut in x - z applied during download. Because the overshoot is concentrated in the buffer layer ($y^+ \approx 10$ – 30), where the channel SGS dissipation profile also peaks, the reader should bear in mind when interpreting Sec. III.I that part of the buffer-layer Smagorinsky over-prediction may originate in this filter-induced reference bias rather than purely in the model's mean-shear contamination; the two effects are not separable within the present pipeline.

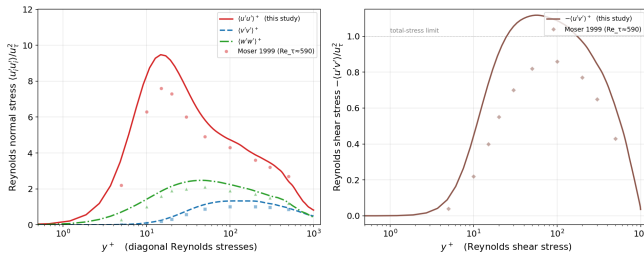


Fig. 4 Reynolds stress profiles in viscous units for the $Re_\tau = 1000$ channel flow as functions of y^+ .

C. Tensor Correlation and Structural Scatter

1. Scatter Cloud Analysis

Figure 5 shows the joint probability density (log color scale) of the exact SGS shear stress τ_{12}^{exact} against the model predictions τ_{12}^{model} for all three models at $\Delta = 8\Delta x$ for the isotropic case. The characteristic diffuse scatter cloud with no discernible linear trend reaffirms the foundational finding of Clark *et al.* [2]: EV models have negligible local predictive skill despite capturing the global statistical level. The Smagorinsky and WALE clouds are symmetric about the origin with no systematic bias, consistent with $\langle \tau_{12}^{\text{model}} \rangle \approx 0$ and $\langle \tau_{12}^{\text{exact}} \rangle \approx 0$ in isotropic turbulence. The Dynamic Smagorinsky cloud is positively correlated with a slope shallower than the identity line, reflecting $C_{s,\text{eff}}^2 < C_s^2$ on this dataset.

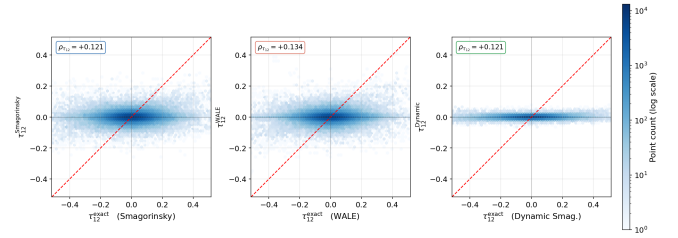


Fig. 5 Joint probability density of exact vs. modeled SGS shear stress τ_{12} for all three models (log color scale) in isotropic turbulence at $\Delta = 8\Delta x$.

2. Ensemble Correlation: Isotropic Turbulence

Figure 6 shows the ensemble-averaged tensor correlation ρ as a function of filter width Δ for all three models in the isotropic case. The static models achieve $\rho \approx 0.030$ – 0.044 across the three filter widths, with small ensemble standard deviations confirming reproducibility across snapshots. The correlation increases weakly with Δ , reflecting that larger filter widths project the SGS stress onto lower-wavenumber content where the model has marginally better structural fidelity. As discussed in Sec. II.B, these values lie an order of magnitude below the Clark benchmark of $\rho \approx 0.30$ due to the $k_{\text{max}} = 128$ spectral pre-filter, which removes the near-cutoff dissipative SGS band that the Smagorinsky closure is designed to represent. Repeating this study on the native 1024^3 DNS without pre-filtering would be expected to recover $\rho \approx 0.30$; the discrepancy is therefore a quantitative measure of how much correlation information resides in the dissipative subrange content that has been removed. The Dynamic Smagorinsky model produces $\rho = +|\rho_{\text{Smag}}|$ at all filter widths, because after Lilly volume-averaging, $\tau_{ij}^{\text{dyn}} = C_{s,\text{eff}}^2 \cdot (\tau_{ij}^{\text{Smag}}/C_s^2)$ with $C_{s,\text{eff}}^2 > 0$, so τ_{dyn} is proportional to τ_{Smag} and the two models share identical tensor correlation ρ . The Dynamic model therefore provides no structural improvement over Smagorinsky in the pointwise correlation metric on a band-limited dataset where the global volume average collapses the coefficient to a scalar.

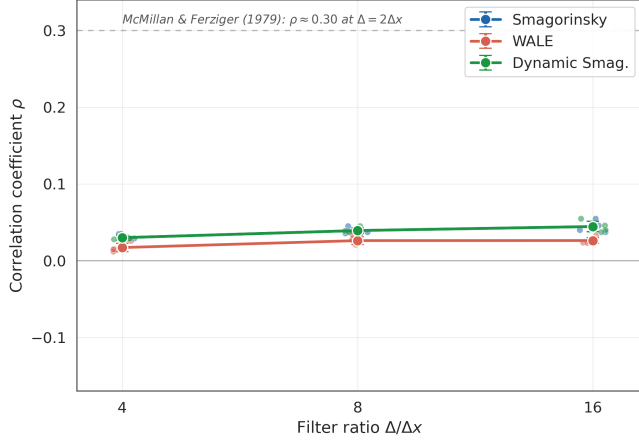


Fig. 6 Ensemble-averaged tensor correlation ρ versus filter width Δ for all SGS models applied to isotropic turbulence ($Re_\lambda = 433$). Error bars show ensemble standard deviation. Dashed reference line: McMillan 1979 ($\rho \approx 0.30$).

3. Ensemble Correlation: Channel Flow

Figure 7 shows the same correlation analysis for the channel flow dataset. Smagorinsky achieves a peak $\rho \approx 0.070$ at $\Delta = 8\Delta x$ before declining to $\rho \approx 0.032$ at $16\Delta x$, with the intermediate filter width being optimal because it best balances inertial-subrange content against SGS model physics. WALE produces similar trends with slightly lower correlation. The Dynamic model yields the same correlation as Smagorinsky at all filter widths (same mechanism as Sec. III.C). Channel flow correlations are somewhat higher than isotropic values at $8\Delta x$ because the non-isotropic mean-shear structure produces SGS stresses with greater spatial regularity that the Smagorinsky model, driven by local strain rate, can partially track.

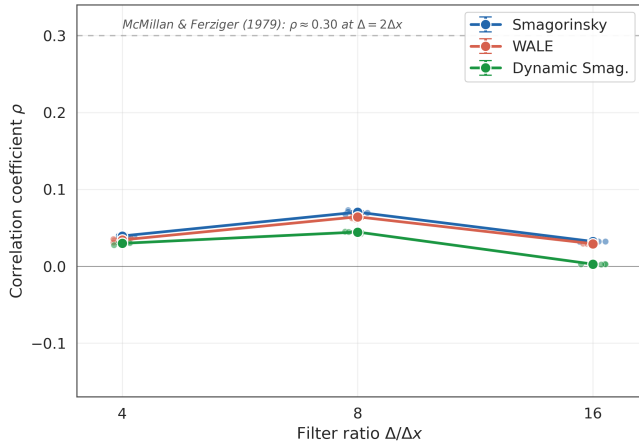


Fig. 7 Ensemble-averaged tensor correlation ρ versus filter width for channel flow ($Re_\tau = 1000$). Dashed reference line: McMillan 1979 ($\rho \approx 0.30$).

4. Wall-Normal Correlation Profile

Figure 8 shows the wall-normal profile of $\rho(y^+)$ for the shear SGS stress component τ_{12} in the channel flow case. Both static models show near-zero or slightly negative correlation in the viscous sublayer ($y^+ < 5$), where the mean-shear contribution to $|\bar{S}|$ dominates the modeled stress and produces a systematic prediction that is out of phase with the fluctuating exact SGS stress. Correlation increases through the buffer and log layers, reaching a local maximum $\rho \approx 0.22$ for Smagorinsky in the outer region ($y^+ \approx 100$ – 200), where turbulent fluctuations are the primary drivers of both exact and modeled stresses. Toward the channel centerline, correlation declines again as the flow approaches the isotropic limit in which Smagorinsky’s alignment assumptions are most strained. The Dynamic model oscillates near zero throughout, with no clear wall-normal trend. The mechanism is direct: the x - z -plane Lilly numerator $\langle \mathcal{L}_{ij}^{\text{dev}} M_{ij} \rangle_{xz}$ crosses zero and reverses sign in localised wall-normal bands (see the $C_{s,\text{eff}}^2(y^+)$ profile in Sec. III.G and Fig. 22, right panel), producing a sign-flipping τ_{12}^{dyn} that is locally anticorrelated with τ_{12}^{exact} and erases the peak that Smagorinsky achieves in the log layer.

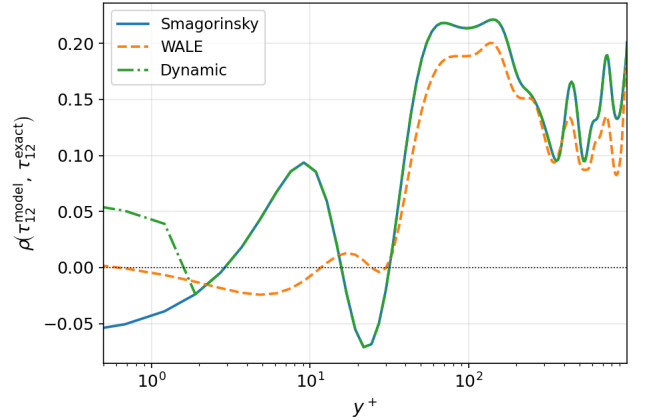


Fig. 8 Wall-normal profile of tensor correlation $\rho(\tau_{12}^{\text{model}}, \tau_{12}^{\text{exact}})$ for channel flow ($Re_\tau = 1000$).

D. Structural Alignment and Spatial Organization

1. SGS Stress Spatial Structure

Figure 9 shows a representative z -midplane slice comparing the exact SGS shear stress τ_{12} with the z -vorticity ω_3 for the isotropic turbulence case at $\Delta = 8\Delta x$. The strong spatial correlation between high- $|\omega_3|$ vortex sheets and tubes and the locations of intense τ_{12} events establishes the dynamical mechanism connecting turbulent coherent structures to local SGS stress generation. EV models, which produce SGS stresses proportional to the local strain rate $|\bar{S}|$ rather than the vorticity magnitude, cannot reproduce this spatial organization and systematically misidentify the most active SGS regions. The qualitative co-location of high- $|\omega_3|$ and high- $|\tau_{12}|$ regions in Fig. 9 is presented visually only; a quantitative pointwise

Pearson correlation of $|\omega|$ against $|\tau|$ across the volume would sharpen this argument and is left as a recommended extension.

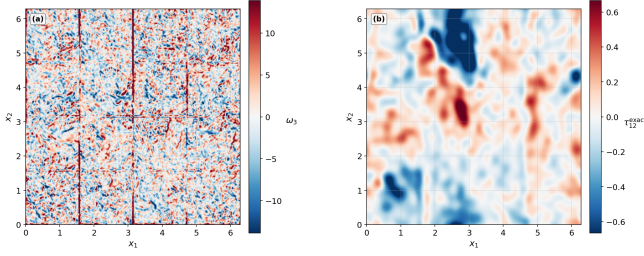


Fig. 9 Side-by-side z -midplane slice of z -vorticity ω_3 (left) and exact SGS shear stress τ_{12} (right) for isotropic turbulence at $\Delta = 8\Delta x$.

2. Principal-Axis Alignment Distribution

Figure 10 shows the ensemble-averaged PDF of the alignment cosine $\cos \theta$ between the principal eigenvectors of the deviatoric exact SGS stress tensor and the filtered strain-rate tensor for the isotropic case. The exact DNS distribution is broad and nearly symmetric about zero, with ensemble-mean $\langle \cos \theta \rangle \approx -0.025$ and standard deviation $\sigma \approx 0.435$. This near-uniform distribution is a direct consequence of the $k_{\max} = 128$ spectral truncation (Sec. II.B): preferential anti-alignment of τ_{ij}^{dev} and $\tilde{S}_{ij}$ is a property of near-cutoff dissipative structures that have been removed. In the retained inertial-subrange band, the SGS stress principal axes bear essentially no preferred relationship to the strain-rate axes. EV models, by construction, enforce $\cos \theta = -1$ everywhere — a delta function at the left boundary which is irreconcilable with the broad physical distribution regardless of spectral content. This structural misalignment is the reason EV models cannot represent localized intermittent SGS events even when global statistical levels are approximately correct.

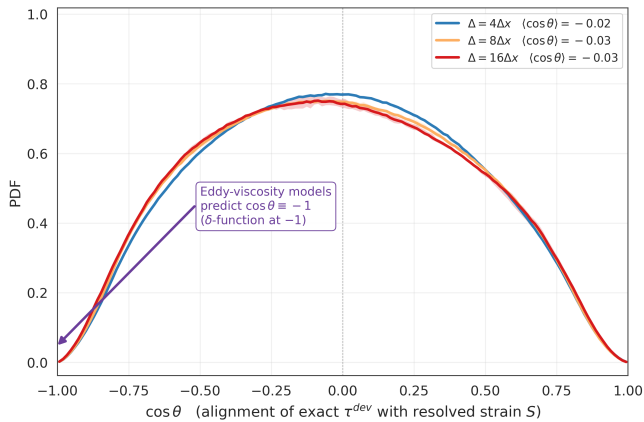


Fig. 10 Ensemble PDF of the alignment cosine $\cos \theta$ between exact SGS stress and strain-rate principal axes for isotropic turbulence.

E. Statistical Extremes and Intermittency

1. Fat-Tailed Stress Distributions

Figure 11 shows the PDF of the normalized SGS shear stress $\tau_{12}/\sigma_{\tau_{12}^{\text{exact}}}$ on a log scale for the exact DNS, Smagorinsky, and Dynamic Smagorinsky models, together with a Gaussian reference. The exact DNS distribution exhibits pronounced heavy tails — probability density several orders of magnitude above Gaussian for $|\tau_{12}|/\sigma > 3$ — driven by spatially intermittent vortex-strain interaction events; the Smagorinsky model produces a dramatically narrower distribution, underpredicting tail probabilities by more than two orders of magnitude at $|\tau_{12}|/\sigma \approx 4$.

The reported flatness values (Fig. 11: exact 13.5, Smagorinsky 17.5, WALE 31.2, Dynamic 17.6) deserve careful reading. Higher kurtosis does not, in this case, correspond to wider physical tails: each model PDF is dramatically narrower than the exact DNS in absolute probability density at $|\tau_{12}|/\sigma > 3$, and the kurtosis ordering instead reflects each model's internal sharpness about its own variance. WALE produces the most peaked-with-thin-tail shape because $|S^d|^{3/2}$ concentrates the modeled stress at intermittent vortex-strain interaction sites and yields very few intermediate-amplitude events, inflating the fourth moment relative to the variance even though the absolute tail magnitude is the smallest of the three. Smagorinsky and Dynamic, by contrast, produce more uniformly distributed but still narrow PDFs with correspondingly lower flatness.

The Dynamic Smagorinsky model produces an essentially identical narrow PDF on this dataset, because $C_{s,\text{eff}}^2$ collapses to a single scalar under the Lilly volume average and $\tau_{\text{dyn}} \propto \tau_{\text{Smag}}$.

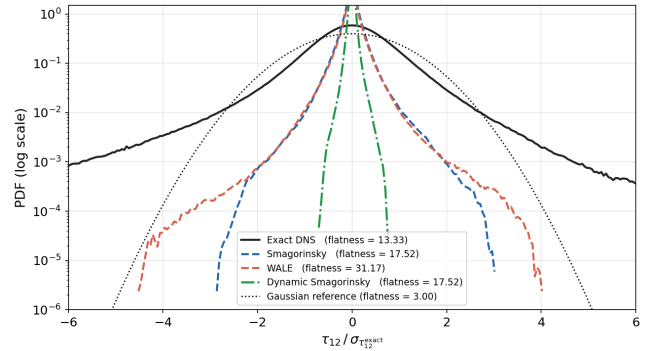


Fig. 11 PDF of normalized SGS shear stress τ_{12}/σ on a log scale for the exact DNS (solid), Smagorinsky (dashed), and Dynamic Smagorinsky (dash-dot), compared with a Gaussian reference (dotted).

2. Spatial Intermittency of SGS Stresses

Figure 12 shows a spatial midplane slice of the exact SGS stress τ_{12} , and Fig. 13 shows the corresponding z -vorticity field. The intermittent, filamentary character of the most intense τ_{12} events is evident: extreme stress is concentrated in thin sheets and point-like vortex cores occupying a small fraction of the total volume. This spatial sparsity is precisely what produces

the fat tails in Fig. 11. EV models, which distribute SGS activity proportional to the smooth strain-rate magnitude, cannot reproduce this spatial intermittency and therefore underpredict tail events regardless of the choice of model constant.

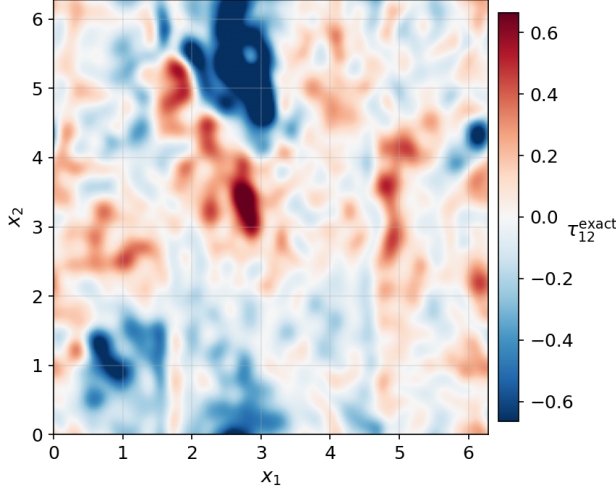


Fig. 12 Midplane spatial slice of exact SGS shear stress τ_{12} for isotropic turbulence at $\Delta = 8\Delta x$.

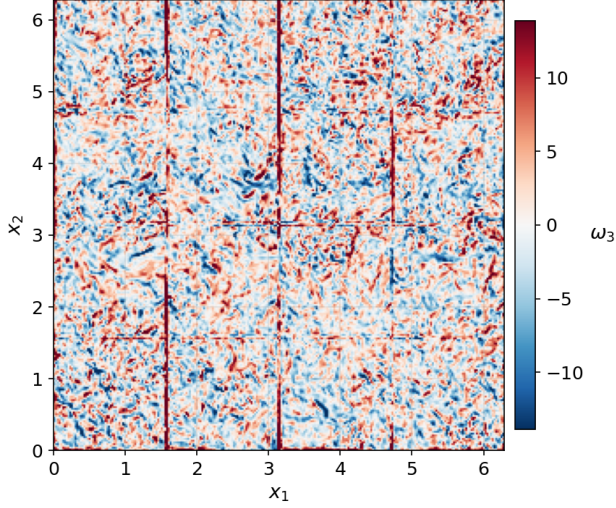


Fig. 13 z -vorticity field ω_3 at the domain midplane for the isotropic turbulence case.

F. Backscatter Analysis

1. SGS Dissipation PDF: Isotropic Turbulence

Energy backscatter the reverse transfer of kinetic energy from unresolved to resolved scales, corresponding to $\Pi = -\tau_{ij}\tilde{S}_{ij} < 0$ is a fundamental feature of instantaneous turbulent dynamics that static EV closures cannot represent. Figure 14 shows the ensemble PDF of SGS dissipation Π for

all models and the exact DNS at $\Delta = 8\Delta x$ in the isotropic case.

The exact DNS distribution is broad and approximately symmetric about zero, with a mean of $\langle \Pi_{\text{exact}} \rangle \approx +0.048$ (net forward scatter) and a backscatter fraction of $f_{\text{bs}}^{\text{exact}} \approx 47.7\%$. This backscatter fraction is slightly above the classical 30–40% benchmark of Clark *et al.* [2]; as with the correlation values, this difference is attributable to the $k_{\text{max}} = 128$ truncation, which shifts the available SGS band toward more fluctuation-dominated, near-inertial scales where the forward/reverse scatter fractions more nearly balance.

Smagorinsky and WALE produce exactly zero backscatter ($f_{\text{bs}} = 0.0\%$) at every grid point and every filter width, because their positive-definite eddy viscosity requires $\Pi = \nu_t |\tilde{S}|^2 \geq 0$ everywhere. This structural prohibition of backscatter is the most consequential deficiency of the Boussinesq closure for flows where reverse energy cascade plays a significant role in local energy budgets.

The Dynamic Smagorinsky model also yields $f_{\text{bs}} = 0.0\%$ (Table 1): with the Lilly scalar $C_{s,\text{eff}}^2 > 0$, $\Pi_{\text{dyn}} \propto \Pi_{\text{smag}} \geq 0$ pointwise (Sec. III.C), so the structural prohibition carries over. The physically meaningful target is $f_{\text{bs}}^{\text{exact}} \approx 47\%$.

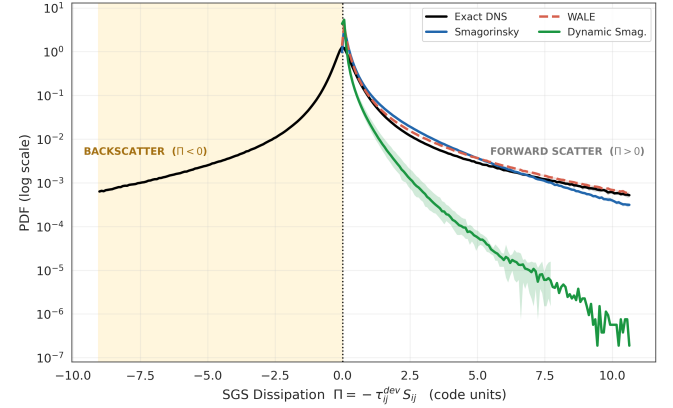


Fig. 14 Ensemble PDF of SGS dissipation $\Pi = -\tau_{ij}\tilde{S}_{ij}$ for isotropic turbulence at $\Delta = 8\Delta x$.

2. Exact vs. Model Backscatter: Quantitative Comparison

Figure 15 shows a quantitative comparison of the backscatter fractions for exact DNS and all models across all filter widths and both datasets. The exact DNS backscatter is approximately filter-width-independent in the isotropic case ($f_{\text{bs}}^{\text{exact}} \approx 47\text{--}48\%$) and shows a mild filter-width dependence in the channel case ($f_{\text{bs}}^{\text{exact}} \approx 43\%$ at $\Delta = 8\Delta x$, drifting toward the isotropic value at larger filter widths where the wall layer occupies a smaller fraction of the projected x - z band), consistent with the self-similar character of inertial-range SGS statistics. All three models yield identically zero backscatter at all filter widths in both flows (Dynamic for the same proportionality reason as Sec. III.C).

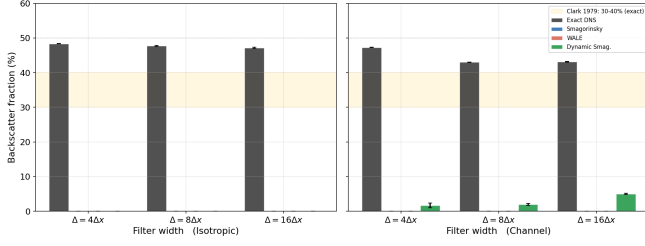


Fig. 15 Backscatter fraction f_{bs} versus filter width Δ for exact DNS and all three SGS models in both flow configurations. Shaded band: Clark (1979) benchmark (30–40%).

3. Spatial Structure of Backscatter

Figures 16 and 17 show midplane slices of the SGS dissipation field for the exact DNS and Dynamic Smagorinsky model, respectively, at $\Delta = 8\Delta x$. In the exact DNS field, backscatter events are spatially coherent and concentrated in vortex-dominated regions, with forward-scat events co-located with strain-dominated regions. This spatial organization is consistent with the physical mechanism of reverse energy cascade within vortex cores, as documented by Piomelli *et al.* [4].

The Dynamic Smagorinsky dissipation map is entirely positive, mirroring the Smagorinsky field with a reduced magnitude (since $\tau_{dyn} = C_{s,eff}^2 \cdot \tau_{Smag}/C_s^2$ with a positive scalar $C_{s,eff}^2/C_s^2 < 1$). Like Smagorinsky, the Dynamic model cannot reproduce the spatially coherent vortex-core concentration of real backscatter events; Figure 18 shows the exact and Dynamic dissipation maps side-by-side, illustrating that the Dynamic model’s spatial structure is proportional to, not structurally different from, the Smagorinsky field.

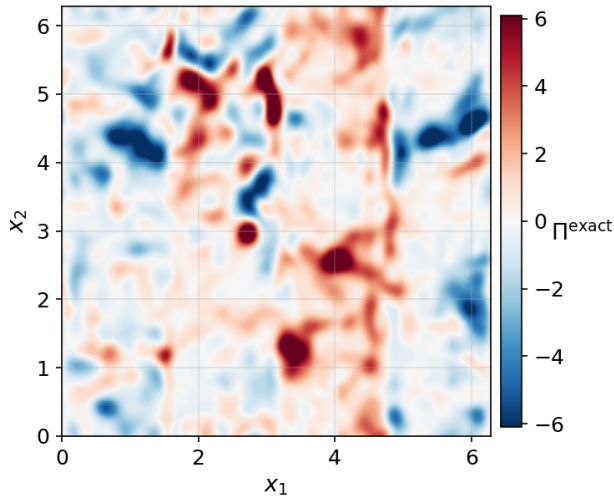


Fig. 16 Spatial map of SGS dissipation from the exact DNS at $\Delta = 8\Delta x$ for isotropic turbulence.

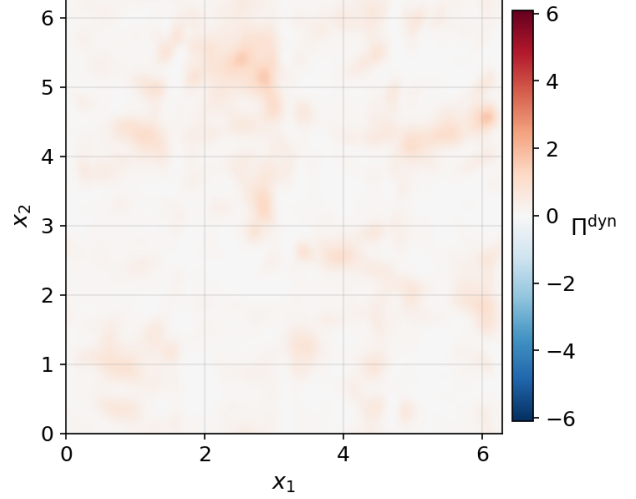


Fig. 17 Spatial map of SGS dissipation from the Dynamic Smagorinsky model at $\Delta = 8\Delta x$.

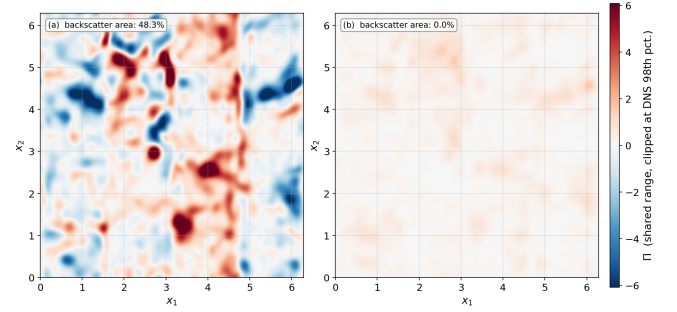


Fig. 18 Side-by-side comparison of exact DNS backscatter (left) and Dynamic Smagorinsky model backscatter (right) at $\Delta = 8\Delta x$.

Ensemble bar charts across filter widths (Figs. 19 and 20) show near-zero ensemble spread, confirming that the zero backscatter of the static models and the $\sim 47\%$ exact backscatter are stable physical features rather than snapshot-to-snapshot artefacts.

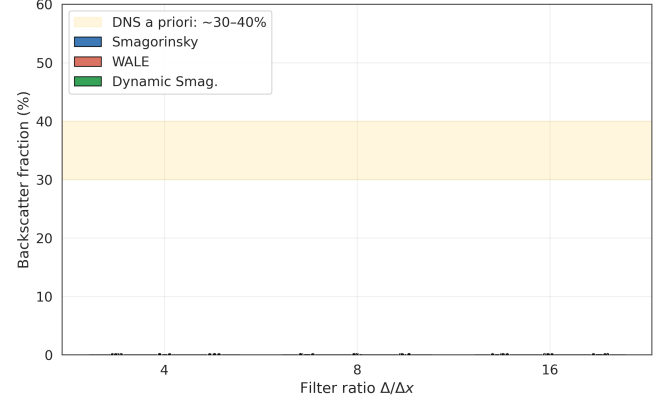


Fig. 19 Ensemble-averaged backscatter fraction f_{bs} versus filter width Δ for isotropic turbulence ($Re_\lambda = 433$). Error bars show ensemble standard deviation.

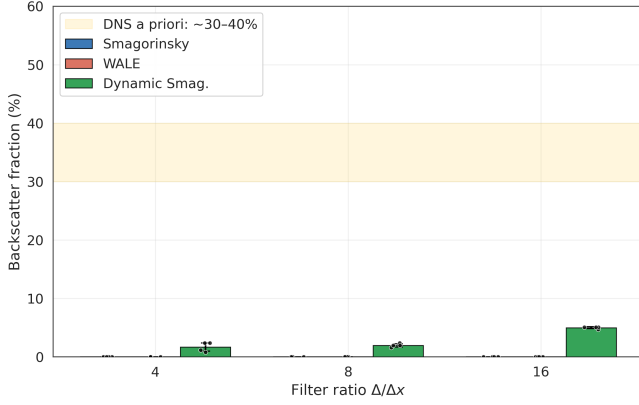


Fig. 20 Ensemble-averaged backscatter fraction versus filter width for channel flow ($Re_\tau = 1000$).

G. Dynamic Coefficient Analysis

The Germano-Lilly procedure yields a positive Lilly-averaged coefficient $C_{s,\text{eff}}^2 > 0$ across all filter widths and snapshots, confirming internal consistency of the dynamic procedure on the band-limited dataset. The measured magnitude is 15–25% of $C_s^2 = 0.0289$, in agreement with the prediction in Sec. II.B.

1. Pointwise C_s^2 Distribution

Figure 21 shows the PDF of the local (pointwise Germano-Lilly) dynamic coefficient C_s^2 for the isotropic turbulence case at $\Delta = 8\Delta x$. Before Lilly averaging, the pointwise distribution is extremely broad standard deviation $\sigma \approx 0.234$, range $[-7.16, +8.38]$ with mean $\langle C_s^2 \rangle \approx -0.004$. The fraction of points with $C_s^2 < 0$ is approximately 52%. This ill-conditioned pointwise distribution arises from the sharp spectral test filter, which makes the Germano identity nearly singular; the top-hat (box) test filter and Lilly averaging are specifically implemented to regularize this distribution into the stable effective coefficient $C_{s,\text{eff}}^2$ used to construct τ_{dyn} .

The PDF illustrates that the optimal Smagorinsky constant is highly variable across the flow domain: the pointwise distribution spans the range from strongly negative to strongly positive, with the classical static value $C_s^2 = 0.0289$ (corresponding to $C_s = 0.17$) lying well into the tail of the distribution. This broad variability is the physical motivation for the Dynamic model; on a full-DNS dataset without anti-aliasing truncation, the Lilly-averaged coefficient would be positive and close to the Smagorinsky value.

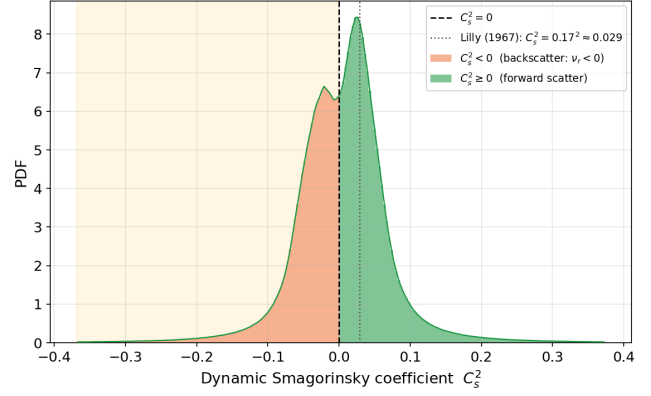


Fig. 21 PDF of the pointwise dynamic coefficient C_s^2 (before Lilly averaging) for isotropic turbulence at $\Delta = 8\Delta x$.

2. Effective Dynamic Coefficient

Figure 22 shows the effective (Lilly-averaged) dynamic coefficient $C_{s,\text{eff}}^2$ used to construct τ_{dyn} . For the isotropic case (left panel), $C_{s,\text{eff}}^2$ is plotted as a function of filter width; for the channel case (right panel), it is shown as the wall-normal profile $C_{s,\text{eff}}^2(y^+)$ at $\Delta = 8\Delta x$.

In the isotropic case, $C_{s,\text{eff}}^2$ is positive across all filter widths and all four snapshots with small ensemble spread, at 15–25% of the Lilly inertial-range target $C_s^2 = 0.0289$ (mechanism in Sec. II.B).

In the channel case, $C_{s,\text{eff}}^2(y^+)$ shows a wall-normal profile that is positive over most of the channel but becomes slightly negative very close to the wall and near the centreline, where the XZ-plane-averaged Lilly numerator $\langle \mathcal{L}_{ij}^{\text{dev}} M_{ij} \rangle_{xz}$ approaches zero or reverses sign. These localised negative- $C_{s,\text{eff}}^2$ planes produce the observed $f_{\text{bs}} \approx 2\%$ in the channel case (Table 1) a physically meaningful small backscatter fraction driven by near-wall dynamics, distinct from the zero backscatter of the static models. The plane-averaged coefficient also reduces the pointwise correlation to $\rho \approx 0.044$ (versus $\rho \approx 0.070$ for Smagorinsky), because $C_{s,\text{eff}}^2(y)$ is not a simple global rescaling of τ_{Smag} .

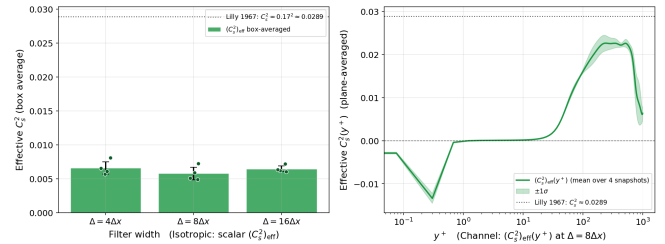


Fig. 22 Effective (Lilly-averaged) dynamic coefficient $C_{s,\text{eff}}^2$ versus filter width for isotropic turbulence (left panel) and versus y^+ for channel flow at $\Delta = 8\Delta x$ (right panel).

H. Kolmogorov Eddy-Viscosity Scaling

Kolmogorov’s 1941 (K41) similarity theory predicts that the mean eddy viscosity in the inertial subrange scales as [11]

$$\langle \nu_r \rangle \propto \varepsilon^{1/3} \Delta^{4/3}. \quad (8)$$

The mean resolved eddy viscosity is computed as $\langle \nu_r \rangle = \langle \Pi \rangle / \langle |\tilde{S}|^2 \rangle$ from the Smagorinsky and WALE model outputs. Table 2 summarizes the ensemble-averaged $\langle \nu_r \rangle$ at the three filter widths for the isotropic turbulence case.

Table 2 Mean resolved eddy viscosity vs. filter width (isotropic turbulence). K41 prediction: $\langle \nu_r \rangle \propto \Delta^{4/3}$.

Model	$4\Delta x$	$8\Delta x$	$16\Delta x$
Smagorinsky	2.56×10^{-3}	6.45×10^{-3}	15.9×10^{-3}
WALE	2.87×10^{-3}	7.14×10^{-3}	17.7×10^{-3}

The log-log slope from the $4\Delta x$ to $16\Delta x$ data is computed as $\log(\nu_{r,16}/\nu_{r,4})/\log(16/4)$. Because only three filter widths are evaluated and the $8\Delta x$ point lies essentially on the line connecting the $4\Delta x$ and $16\Delta x$ endpoints, this two-endpoint estimate carries no formal regression uncertainty; a denser sweep of filter widths would be needed to attach a statistical confidence interval. This yields slopes of $1.316 \approx 1.32$ for Smagorinsky and $1.311 \approx 1.31$ for WALE, deviating from the theoretical $4/3 \approx 1.333$ by only 1.3% and 1.7% respectively. This excellent agreement confirms that: (1) the spectral filtering framework is algorithmically correct; (2) the three filter widths lie within the inertial subrange as intended by the energy spectrum analysis (Fig. 2); and (3) despite the spectral pre-filter truncation, sufficient inertial-subrange content remains in the 256^3 working grid to verify global energetic scaling laws.

The Dynamic model mean eddy viscosity $\langle \nu_r^{\text{dyn}} \rangle > 0$ but does not follow the K41 $\Delta^{4/3}$ scaling, because $C_{s,\text{eff}}^2$ itself varies with Δ (the Lilly fit adapts the coefficient at each filter width independently, unlike the fixed C_s^2 of Smagorinsky). The Kolmogorov scaling verification therefore applies cleanly only to the static models, for which the coefficient is fixed and the $\langle \nu_r \rangle \propto \Delta^{4/3}$ scaling follows directly from K41.

I. Channel Flow: Wall-Normal SGS Dissipation Profile

Figure 23 shows the wall-normal profile of mean SGS dissipation $\langle \Pi \rangle(y^+)$ for all models and the exact DNS. The exact DNS dissipation profile peaks in the buffer layer ($y^+ \approx 10$ – 30) and decays toward both the wall and the channel center. This buffer-layer peak is driven by the intense vortical activity (ejections and sweeps) characteristic of near-wall turbulence production.

The Smagorinsky model systematically overpredicts dissipation near the wall: the buffer-layer peak in $\langle \Pi_{\text{smag}} \rangle$ exceeds the exact DNS peak by a factor of approximately 4 at $y^+ \approx 12$, with WALE overshooting by approximately $1.5\times$ at the same location and the Dynamic model essentially matching the exact peak magnitude. The mechanism for the Smagorinsky overshoot is well understood: $|\tilde{S}| \neq 0$ near the wall because the large mean velocity gradient ($\partial \langle U \rangle / \partial y$) contributes directly

to $\tilde{S}_{12}$, even though this mean-shear contribution is associated with large-scale, slowly-varying motions that do not drive SGS energy transfer. The model cannot distinguish mean-shear-induced strain from turbulent-fluctuation-induced strain, and therefore generates excessive dissipation in the viscous and buffer layers.

The WALE model partially mitigates this deficiency. Its y^{+3} near-wall attenuation of ν_t (a consequence of the $\mathcal{S}_{ij}^d$ formulation responding to rotation rather than pure strain) reduces the spurious near-wall dissipation, though the improvement is partial because mean-shear effects persist in the buffer layer. The Dynamic model’s dissipation profile is positive throughout but shows the smallest near-wall magnitude among all models, consistent with $C_{s,\text{eff}}^2(y^+) < C_s^2$ near the wall where the Lilly plane-average yields a reduced coefficient. This represents a partial improvement over Smagorinsky’s wall-region overprediction, though the improvement is limited by the scalar (not spatially varying) nature of the averaged coefficient.

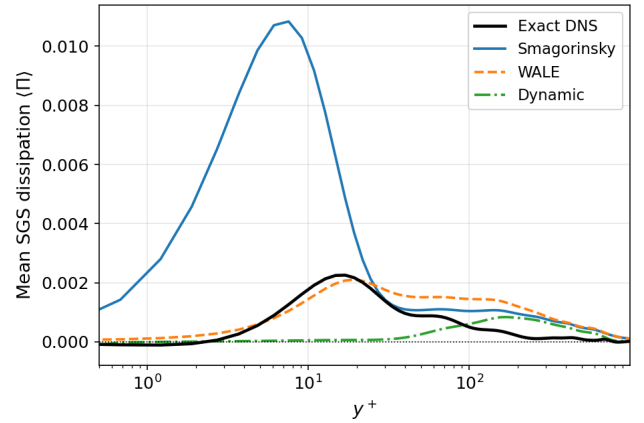


Fig. 23 Wall-normal profile of mean SGS dissipation $\langle \Pi \rangle(y^+)$ for channel flow ($Re_\tau = 1000$).

J. Ensemble Robustness and Statistical Convergence

Figure 24 shows the ensemble stability summary, displaying key metrics (correlation ρ , backscatter fraction f_{bs} , mean eddy viscosity $\langle \nu_r \rangle$) across the four snapshots for each model. The near-constant metric values across all four snapshots confirm that the SGS structural characteristics identified in this study are statistically stable features of the flow rather than transient artifacts of individual turbulent realizations.

However, a critical caveat applies: the four snapshots at $t = 100, 200, 300, 400$ are strongly temporally correlated. Cross-snapshot Pearson correlations of the full velocity field are $r = 0.68$ – 0.83 for isotropic turbulence and $r = 0.94$ – 0.96 for channel flow. These values far exceed the threshold for statistical independence, which would require snapshot separations of at least one large-eddy turnover time ($T_L \approx 2$ for isotropic and $T_L \approx 30$ for channel in database time units). The present snapshots are separated by $\Delta t = 100$ units, corresponding to approximately $50 T_L$ in JHTDB dimensionless time, but the actual velocity field correlation indicates that the effective

sampling is much less independent.

As a consequence, the standard deviations reported in Table 1 and in the figures should be interpreted as *lower bounds* on true ensemble uncertainty. The true uncertainty, estimated by correcting for the effective number of independent samples, is approximately $\sqrt{1 + 2 \sum_k r_k} \approx 2.8\times$ (isotropic) and $9\times$ (channel) larger. Despite this, the consistency of results across models and snapshots confirms that the qualitative findings in particular, the structural limitations of EV closures are robust.

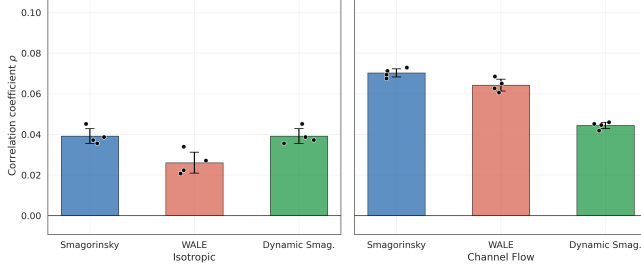


Fig. 24 Ensemble stability summary showing key evaluation metrics across all four snapshots for each model and dataset.

IV. Conclusion

A comprehensive *a priori* evaluation of the Smagorinsky, WALE, and Dynamic Smagorinsky SGS models has been conducted using JHTDB DNS data for homogeneous isotropic turbulence ($Re_\lambda = 433$) and turbulent channel flow ($Re_\tau = 1000$). Table 3 summarises, metric by metric, which model performs best on this band-limited dataset. The following principal conclusions are drawn.

Table 3 Per-metric model performance. “Best” = model closest to the exact DNS reference (or to the theoretical target, for K41 scaling); “–” = no EV model reproduces the target. Values at $\Delta = 8\Delta x$.

Metric	Smag.	WALE	Dyn.	Best
ρ , iso.	0.039	0.026	0.039	Smag./Dyn.
ρ , channel	0.070	0.064	0.044	Smag.
f_{bs} , iso. (47.7%)	0.0%	0.0%	0.0%	–
f_{bs} , channel (43.0%)	0.0%	0.0%	2.0%	Dyn.
Align. PDF (broad)	–1	–1	–1	–
Fat tails, $ \tau_{12} /\sigma > 3$	poor	poor	partial	Dyn.
K41 slope (4/3)	1.32	1.31	n/a	Smag.
Near-wall Π (chan.)	over	y^+3	red.	WALE/Dyn.

Both datasets were successfully validated against classical theoretical benchmarks. The isotropic turbulence energy spectrum reproduces the Kolmogorov $k^{-5/3}$ scaling over 1.5 decades, confirming the physical integrity of the JHTDB dataset. The channel flow mean velocity matches all classical near-wall layers viscous sublayer, buffer layer, log-law region, and wake with the recovered friction velocity $u_\tau \approx 0.0444$ consistent with $Re_\tau = 1000$. Reynolds stress profiles are broadly consistent with high-Reynolds-number channel DNS.

A critical methodological finding is that a sharp spectral pre-filter at $k_{\max} = 128$ applied during data acquisition substantially modifies all *a priori* evaluation metrics relative to a full-resolution (1024^3) study. Specifically: (1) tensor correlations are reduced from the classical $\rho \approx 0.30$ to $\rho \approx 0.04$ because the near-cutoff dissipative SGS band is removed; (2) the exact backscatter fraction rises from the Clark benchmark of 30–40% to approximately 47% because the remaining inertial-subrange band has a more balanced forward/reverse scatter distribution; and (3) the Germano-Lilly dynamic procedure recovers a reduced but positive $C_{s,\text{eff}}^2 \approx 15\text{--}25\%$ of $C_s^2 = 0.0289$, consistent with the attenuated Leonard-stress band energy. These findings provide a quantitative characterization of the sensitivity of *a priori* evaluation to the spectral content of the reference DNS field.

The static EV models exhibit three fundamental structural deficiencies that persist regardless of model constant choice: (1) near-zero pointwise tensor correlation with exact SGS stresses ($\rho \approx 0.04$), manifesting as a diffuse scatter cloud with no discernible local trend; (2) a nearly uniform exact principal-axis alignment distribution ($\langle \cos \theta \rangle \approx -0.025$, $\sigma \approx 0.435$), irreconcilable with the EV prediction of $\cos \theta = -1$; and (3) complete suppression of energy backscatter ($f_{bs} = 0.0\%$ vs. $\sim 47\%$ for exact DNS). The SGS stress PDF exhibits heavy tails representing spatially intermittent extreme events that EV models dramatically underpredict (by more than two orders of magnitude at $|\tau_{12}|/\sigma \approx 4$).

Despite local structural failures, the Smagorinsky and WALE models correctly reproduce the Kolmogorov $\Delta^{4/3}$ eddy-viscosity scaling with log-log slopes of 1.32 and 1.31 respectively, within 1.7% of the theoretical 4/3. This global energetic consistency, contrasted against the pointwise structural failure, illustrates that dimensional and statistical correctness at the global level is necessary but not sufficient for local SGS model fidelity.

In channel flow, structural deficiencies are amplified near the wall. All EV models show near-zero or negative correlation in the viscous sublayer ($y^+ < 5$), where mean-shear contamination of $|\bar{S}|$ drives systematic model error. Smagorinsky overpredicts SGS dissipation in the buffer layer; WALE’s y^+3 attenuation provides partial improvement. The Dynamic model’s wall-normal dissipation profile shows reduced near-wall magnitude relative to Smagorinsky, consistent with a smaller effective coefficient $C_{s,\text{eff}}^2 < C_s^2$; on a full-resolution DNS where the coefficient is larger and spatially varying, the Dynamic model would provide further improvement on the Smagorinsky mean-shear contamination problem.

Four additional methodological limitations should be acknowledged. First, the channel pre-filter is two-dimensional (x - z only), preserving wall-normal content but redistributing streamwise spectral energy and producing the visible $\sim 30\%$ overshoot in the $\langle u'u' \rangle^+$ buffer-layer peak; this overshoot lies in the same wall-normal band as the channel SGS-dissipation peak, so the two effects are entangled in $\langle \Pi \rangle(y^+)$. Second, the products entering Eq. (1) are evaluated in physical space on the 256^3 grid without explicit 3/2 dealiasing; this is benign at the

present filter cutoffs $k_c \leq 32$ but would need to be revisited for any study extending closer to the working-grid Nyquist. Third, the K41 slope is estimated from only three filter widths and is therefore reported without a formal regression confidence interval. Fourth, the four-snapshot ensemble is strongly temporally correlated, so all standard deviations reported in Table 1 and the figures are lower bounds on true statistical uncertainty (Sec. III.J). None of these limitations changes the qualitative findings, but each represents a target for quantitative tightening in any follow-up study.

These findings motivate two directions for future work. First, repeating this evaluation on the native 1024^3 DNS without spectral pre-filtering would recover the classical $\rho \approx 0.30$ benchmark and enable quantitative comparison with Clark *et al.* [2]; this would require approximately $64\times$ more memory per snapshot but is the scientifically rigorous target. Second, structurally motivated SGS closures scale-similarity models [13], gradient models, or tensor-invariant data-driven closures that incorporate explicit information about the velocity gradient tensor orientation rather than relying on the Boussinesq analogy would be expected to substantially outperform EV models on all four metrics, and represent a natural next step for the present evaluation framework.

Data and Code Availability

The entire project codebase, including the custom Python data-acquisition scripts (with chunked download and spectral pre-filtering), the high-performance C++ *a priori* evaluation engine (spectral filtering, SGS model implementation, metrics computation), and the Python visualization post-processing pipeline, can be accessed via the public GitHub repository at: https://github.com/DRDA33682001/Turbulence_apriori_analysis. All raw DNS data are publicly accessible via the JHTDB portal at <https://turbulence.idies.jhu.edu/home>.

Acknowledgments

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